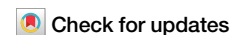


<https://doi.org/10.1038/s44310-024-00029-2>

Realization of high-performance optical metasurfaces over a large area: a review from a design perspective



Minseok Choi^{1,6}, Junkyeong Park^{1,6}, Jehyeon Shin^{2,6}, Harit Keawmuang^{1,6}, Hongyoon Kim¹, Jooyeong Yun¹, Junhwa Seong¹ & Junsuk Rho^{1,3,4,5} ✉

Remarkable advancements have been made in the design of optical metasurfaces in recent years, particularly in compact designs. However, for their practical integration into diverse optical systems, there is a pressing need for metasurfaces to transition toward larger areas without compromising their performance. From a design perspective, efforts in the design process must focus on reducing computational costs and enhancing performance in larger areas. In this review, we introduce diverse optical analyses applicable to wide areas, including the modification of boundary conditions, fast multipole methods, coupled mode theory, and neural network-based approaches. In addition, inverse design methods based on the adjoint method or deep learning, which are suitable for large-scale designs, are described. Numerous fast and accurate simulation methods make it possible to assess optical properties over large areas at a low cost, whereas diverse inverse design methods hold promise for high performance. By concurrently addressing both the essential aspects of designing large-area metasurfaces, we comprehensively discuss various approaches to develop metasurfaces with high performance over expansive regions. Finally, we outline additional challenges and prospects for realizing mass-produced high-performance metasurfaces, unlocking their full potential for optical applications.

Metasurfaces, which are optical devices comprising structures on the nanometer scale and designed to control the wavefront and properties of light, represent a promising avenue for replacing conventional optical components such as refractive optical elements (ROEs). In contrast to ROEs, metasurfaces can manipulate various optical properties, including polarization, amplitude, and phase, within a single component^{1–5}. Furthermore, they contribute to reducing the form factor of traditional bulky systems by replacing conventional optical elements^{6–8}. Despite these advantages, the limited size of metasurfaces at the micrometer scale has hindered their specific practical applications, such as virtual reality (VR)^{9,10}, astronomical imaging¹⁰, bio-imaging systems^{11–13}, and physically unclonable anti-counterfeiting^{14–17}, which require scales ranging from millimeters to centimeters. However, it is difficult to optically calculate and precisely fabricate millions to billions of nanostructures that constitute metasurfaces.

Consequently, the expansion of metasurface size poses challenges in both the design and fabrication processes.

In conventional fabrication processes, electron beam or focused ion beam lithography have been utilized in the fabrication of metasurfaces when the height of meta atoms is fixed. Direct laser writing and two-photo lithography also have been utilized for fabricating various structure of metasurfaces in multi-height^{18–21}. These methods pose no significant challenges when dealing with a small number of nanostructures. However, for fabricating metasurfaces over large areas, conventional methods are impractical with the number of nanostructures from millions to hundreds of millions, making them unsuitable for mass production with nm scale resolution^{22,23}. Fortunately, recent research has explored techniques, such as nano-imprinting and deep ultraviolet photolithography, for large-scale metasurface fabrication and unveiled the possibility of mass production^{24,25}.

¹Department of Mechanical Engineering, Pohang University of Science and Technology (POSTECH), Pohang, 37673, Republic of Korea. ²Graduate School of Artificial Intelligence, Pohang University of Science and Technology (POSTECH), Pohang, 37673, Republic of Korea. ³Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang, 37673, Republic of Korea. ⁴Department of Electrical Engineering, Pohang University of Science and Technology (POSTECH), Pohang, 37673, Republic of Korea. ⁵POSCO-POSTECH-RIST Convergence Research Center for Flat Optics and Metaphotonics, Pohang, 37673, Republic of Korea. ⁶These authors contributed equally: Minseok Choi, Junkyeong Park, Jehyeon Shin, Harit Keawmuang.

✉ e-mail: jsrho@postech.ac.kr

In terms of design processes, as the size of metasurfaces expands, it becomes imperative to reduce the computational costs associated with their design and minimize performance degradation that occurs as the size increases, particularly when aiming for multifunctionality²⁶, such as broadband achromatic focusing⁷ and controlled angular dispersion²⁷. Furthermore, in the case of rapid phase changes and general incident waves (non-plane waves), full-wave simulations of the total structure are necessary to validate the performance; however, this escalates the computational burden, making it almost intractable as the area expands²⁸. For example, although traditional mesh-based numerical analyses, such as the finite difference time domain (FDTD) or finite element method (FEM), effectively capture the behavior of light, scaling up the dimension and size of the calculation region exponentially increases the computational burden²⁹. To address these limitations, accelerated calculation methods are necessary for fast simulations, utilizing various approximations to achieve a cost-effective analysis of light properties. However, excessive or inadequate approximations can compromise the accuracy and performance of metasurfaces. Hence, a balanced technique between speed and accuracy in calculation methods is crucial, and certain methods have been applied for large-scale analyses such as the modification of boundary conditions³⁰, fast multipole method (FMM)³¹, coupled mode theory (CMT)³², and neural network-based approaches³³.

In addition to these fast and accurate simulation methodologies, an inverse design method has been used for high-performance metasurfaces to overcome the performance limitations of forward design and realize multifunctionalities in a large area^{34,35}. Notably, recent advancements in inverse design methods, including the adjoint method and deep learning, have aimed to solve large and complex problems efficiently and synergize with fast simulation techniques in the field of metasurfaces^{9,36}.

Despite these advancements, there is a noticeable gap in the literature concerning reviews that concurrently cover both rapid and accurate simulation methods and inverse design methodologies in large-scale design processes. To overcome the challenges of scaling metasurfaces and facilitate their practical applications, a comprehensive review encompassing effective simulation and optimization methods for large-scale designs is essential. In this paper, we present a comparative analysis of fast calculation methods suitable for large-scale metasurface designs while ensuring reliable accuracy (Section “Fast simulation method”). In the subsequent section, we delve into inverse design methods that are used to realize high-performance

metasurfaces on a large scale compared with traditional forward design approaches (Section “Inverse design method”) (Fig. 1).

Fast simulation method

Modification of boundary conditions

A typical forward design involves importing meta-atoms corresponding to each phase from a library to achieve the desired phase mask. To construct the meta-atom library, a meta-atom in one unit cell is conducted under periodic boundary conditions, and the complex electric field value is stored. The amplitude is determined by the absolute value of the complex field, whereas the phase is represented by the angle value. Once the library is established, the phase function is discretized on a grid with a resolution matching that of the metasurface periodicity. Within a single period, each meta-atom corresponds to a grid phase. At this stage, the electromagnetic response of the metasurface within a single grid is assumed to be periodic in the x and y directions, assuming that the surrounding grid is composed of the same meta-atoms as the corresponding grid. This approximation is known as the unit cell method, local phase approximation, or local periodic approximation (LPA) (Fig. 2a)³⁷. In general, for cases such as periodic waveguide grating or low NA metalens operating at a single wavelength, where the structure of the neighboring meta-atoms does not vary significantly, the LPA assumption does not pose significant issues even in designing large-area metasurfaces. This method considers only phase information, neglecting amplitude and higher diffracted orders. If the size of the unit cell exceeds the subwavelength, the contribution of diffracted orders becomes larger. Therefore, the total metasurface differs more significantly from the sum of each unit cell's results. In this case, LPA is not valid, and to accurately observe the total field, it is necessary to simulate the entire metasurface at once³⁸. In this forward design process, the Fourier modal method³⁹, a numerical technique used to analyze the behavior of electromagnetic waves in periodic structures by decomposing the field into its Fourier components and solving Maxwell's equations, is commonly used.

However, in reality, the meta-atoms in the surrounding grid may have slightly different shapes, leading to variations in the coupling between the meta-atoms compared with the actual values. To address these challenges, the local phase method is proposed as a straightforward solution (Fig. 2b). In the study by Hsu et al.⁴⁰, a supercell comprising nine unit cells, each with a 40° phase difference from the adjacent meta-atom, is created to form a library. The total fields E and H are obtained using commercial software

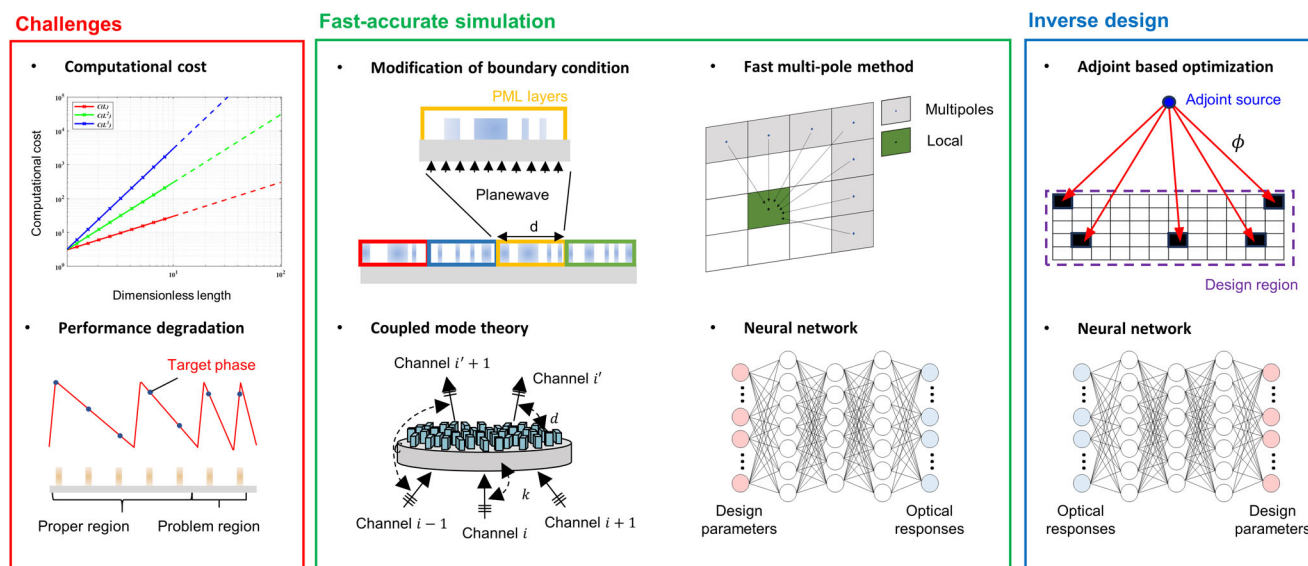


Fig. 1 | Challenges and breakthroughs for large-area metasurface design. Design and simulation of metasurfaces for high performance over large areas entail significant computational costs and bottleneck issues such as inadequate approximations. To address this challenge, there is a need for fast and accurate simulation

methods. By leveraging these methods, it is possible to implement high-performance metasurfaces over large areas through appropriate inverse design methods for optimizing large complexity problems.

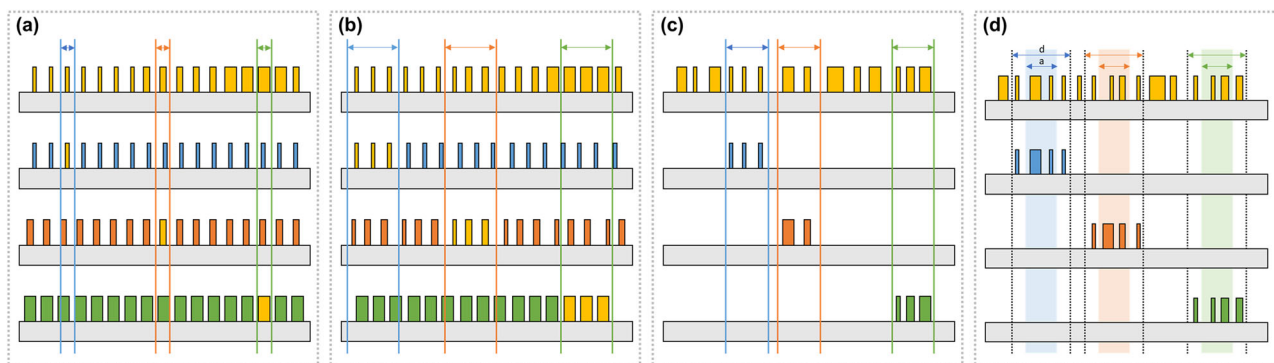


Fig. 2 | Unit cell method and variations by modifying boundary conditions. **a** Unit cell method. The top metasurfaces represent the actual metasurfaces, while the three metasurfaces below depict the assumed surrounding conditions for each unit cell. Although the metasurfaces are actually nonperiodic, the electromagnetic behavior of each unit cell is calculated by presuming that the surrounding unit cells are identical, effectively assuming periodicity³⁷. This method allows for faster metasurface design because it does not simulate the entire area. **b** Increasing the unit cell size can reduce

the incorrect prediction of coupling with surrounding meta-atoms⁴⁰. **c** Modifying the periodic boundary condition to the perfectly matched layer (PML) in inverse design produces more efficient results³⁰. **d** Introducing a spatial overlap region instead of a PML right next to the target cell allows for more efficient simulations (a: basic unit cell and d: area encompassing both the unit cell and overlapping domain regions)⁴⁷.

CST, and equivalent sources (J_s and M_s) are derived using the equivalence principle. Equivalent sources are then employed to record the phases in a new simulation file, thus forming the basis for the library. When a deflector is designed using this library, the radiated energy in the far field or, equivalently, the radar cross section (RCS) performance, is significantly improved by 124%. Furthermore, when applied to the design of a high-numerical aperture (NA) concentrator, the energy at the focal spot is enhanced by 15%.

A previous study showed³⁵ that design methods relying on unit cells and libraries experience limitations in achieving high efficiency and NA. Although topology optimization allows for the creation of devices with improved efficiencies^{35,41}, it demands substantial computational resources, which scale up significantly with the size of the device. Consequently, previous some topology-optimized devices are typically confined to microscale dimensions or to periodic structures composed of microscale unit cells. One way to address this problem is completing a large-area metasurface by stitching individual optimized regions together. To achieve this, the introduction of a perfectly matched layer (PML) instead of periodic boundary conditions has been proposed (Fig. 2c). In the study by Phan et al.³⁰, the Fourier modal method is modified to the aperiodic Fourier modal method (AFMM) by combining the Stratton–Chu integral equations^{42–46}. Research has shown that conducting topology optimization using the AFMM condition within individual sections and then integrating them can result in superior operational efficiency compared with conventional forward designs. This is because, in the single optimized section, this approach considers near-field optical coupling between adjacent nanostructures. Moreover, it is computationally more efficient than performing topology optimization across the entire metasurface. To refine the calculations further, a spatial overlap region is introduced instead of a PML immediately next to it, allowing for more accurate simulations as the overlap region increases (Fig. 2d)⁴⁷. This method is inspired by the Schwarz algorithm and updates the boundary condition in each domain from the neighbor in the next iteration. This approach reduces the error by up to five times while maintaining a similar simulation cost.

Fast multipole method

The FMM is a powerful computational technique employed to accelerate the simulation of complex interactions in various scientific fields^{48,49}. The FMM efficiently computes pairwise interactions among a large number of particles by leveraging hierarchical structures and low-rank approximations to significantly minimize the computational complexity, making it extremely useful for large-scale simulations^{50,51}. The general concept of the FMM involves the calculation of the N-body problem, which is commonly found

in various computational physics scenarios such as molecular simulations^{52–54}, fluid dynamics⁵⁵, and astrophysics⁵⁶. The N-body problem includes a set $X \subset \mathbb{R}^d$ of N target points, a set $Y \subset \mathbb{R}^d$ of N source points, a kernel function $G(x, y)$, and $\{f(y): y \in Y\}$ represents a set of weights at source points where the computation for each $x \in X$ of the potential $u(x)$ is defined as $u(x) = \sum_{y \in Y} G(x, y)f(y)$ ⁵⁷. For example, in electrostatics, the kernel $G(x, y) = 1/|x - y|$ is the Coulomb potential, where X and Y represent the whereabouts of N probes and N charges, respectively⁵⁸. In the case of a direct (naïve) computational method, computing $u(x)$ requires an $O(N^2)$ number of steps, which becomes computationally expensive for a large N ^{59,60}. Moreover, while the direct method provides a very accurate result, it typically requires high memory usage, especially for storing pairwise interaction matrices. To address this concern, the FMM has been introduced to provide an approximate solution with tunable accuracy in only $O(N \log N)$ or $O(N)$ number of steps, depending on the specific implementation and nature of the problem⁶¹. The fundamental concept of the FMM algorithm can be fairly described as follows, as supported by the literatures^{48,49,57}. Given that $X = Y = P$ of N quasi-uniformly distributed points within the unit box $[0, 1]^2$, with $G(x, y) = 1/|x - y|$, the goal is to efficiently compute all pairwise interactions. Consider a problem where A and B are two separated squares of the same size, each containing $O(N)$ points. To approximate pairwise interactions between points in two well-separated squares A and B , sum the masses in B to obtain $f_B = \sum_{y \in B \cap P} f(y)$ and place it at its center c_B , then evaluate the potential $u_A = G(c_A, c_B)f_B$ at the center of A , treating all mass as located at c_B . Finally, use u_A as the approximation for the potential at each point x in square A . This reduces the computation from $O(N^2)$ to $O(N)$ steps, assuming A and B are sufficiently far apart, with a distance greater than or equal to their width. The steps described is a low-rank approximation of the interaction between A and B , and it can be viewed algebraically as follows:

$$G(A, B) \approx \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \cdot G(c_A, c_B) \cdot (1, \dots, 1), \quad (1)$$

While f_A and u_B are just the intermediate outcomes during the application of this low-rank approximation. However, since we are interested in the interaction between all points, this three-step procedure only partially addresses our problem as it only considers the potential in A from points in B . To address interactions between all points, the FMM algorithm uses a quadtree to hierarchically partition the domain until each leaf box contains fewer than a constant $O(1)$ points. After partitioning, the algorithm leverages two key techniques: multipole and local expansions. Multipole

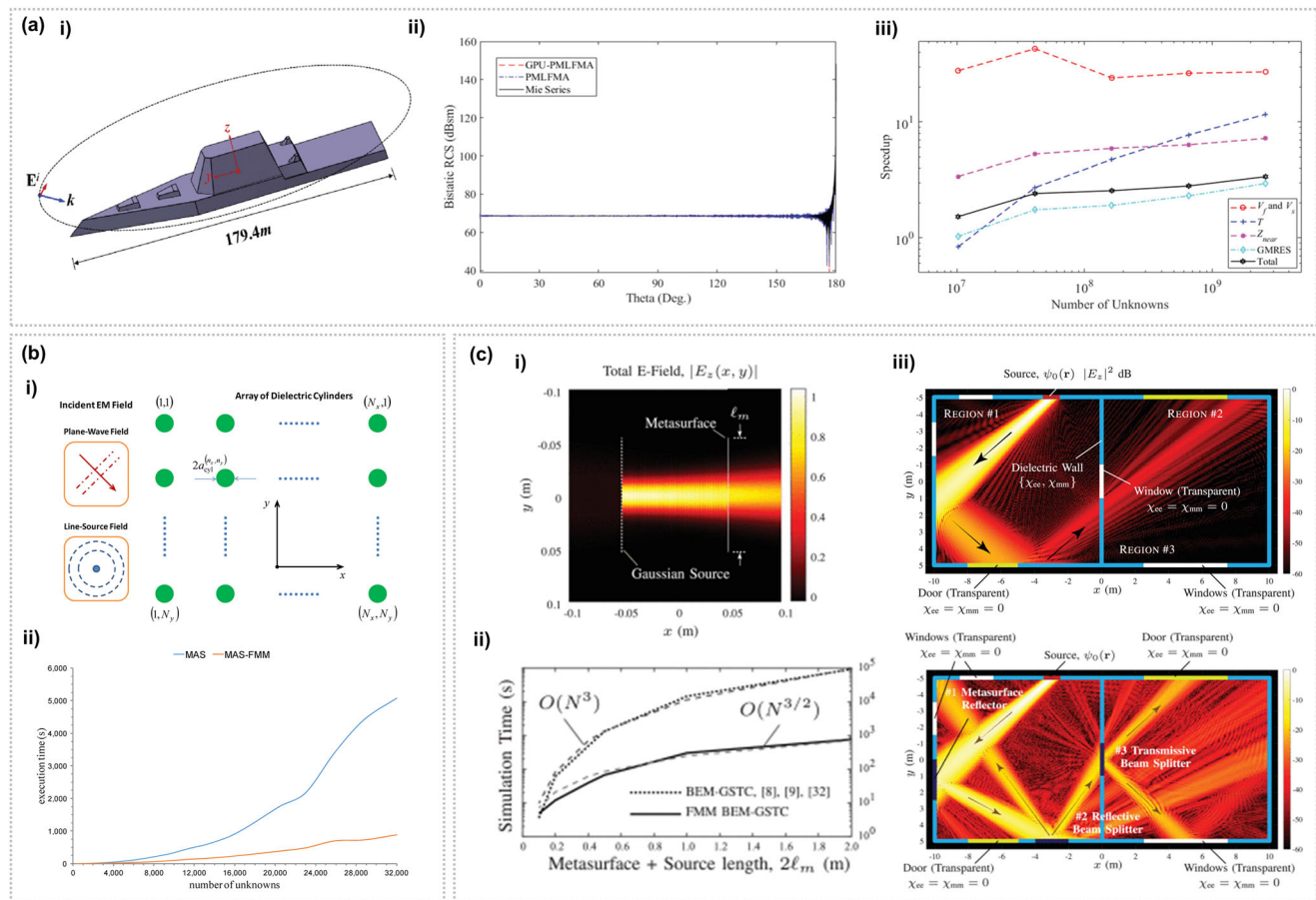


Fig. 3 | Fast multipole method (FMM) implementations in electromagnetic scattering. **a** Parallel graphics processing unit (GPU) and multilevel FMM model for an electrically large and complicated object⁷¹. **a.i** Schematic of the electrically large and complicated object of the ship model. **a.ii** VV-polarized bistatic radar cross-section of the large sphere diameter 3040 m from 0° to 180° generated from the parallel GPU and multilevel FMM (GPU-PMLFMA; red dashed line), PMLFMA (blue dashed line), and Mie series (black solid line). **a.iii** Speedup of the GPU-PMLFMA versus the number of unknowns of the ship model with the aggregation matrix and disaggregation matrix (V_s and V_f ; red dashed line), translation matrix (T ; blue dashed line), near-group impedance matrix (Z_{near} ; magenta dashed line), generalized minimal residual method (GMRES; cyan dashed line), and total speedup (black solid line). **b** Hybridization of the method of auxiliary sources (MAS) with the

FMM⁷⁵. **b.i** Schematic of a large, rectangular array of circular dielectric cylinders, stimulated either by a plane-wave electromagnetic field or an infinite line source running parallel to their axes. **b.ii** Comparison of the required execution times by the conventional MAS method (blue solid line) and the hybridization method (orange solid line) versus the number of unknowns. **c** FMM for large-scale electromagnetic metasurface³¹. **c.i** A two dimensional total E-field when the surface is normally illuminated by a Gaussian beam. **c.ii** Computational time requirements between the standard generalized sheet transition conditions of the integral equations (IE-GSTC; black dashed line) and the FMM IE-GSTC (black solid line) versus the surface length. **c.iii** Schematics of electrically large rooms scenario without (top) and with (bottom) metasurfaces.

expansions are used to approximate the potential field generated by distant groups of sources. Each cluster of points is represented by a multipole expansion at the cluster's center. These expansions are translated and combined hierarchically through the quadtree structure. Local expansions are then used to represent the potential field within a local region, allowing for efficient computation of the potential at nearby points. By converting multipole expansions into local expansions, the FMM efficiently computes interactions with a complexity of $O(N \log N)$ or better.

The FMM can be efficiently employed to solve electromagnetics scattering problems^{62,63}. In the field of electromagnetics, the FMM is commonly enhanced through hybridization with various techniques, such as a hybrid finite element boundary integral, to reduce the computational requirement of the boundary integral with a large contour size from $O(N^2)$ to $O(N^{1.5})$, resulting in a significant reduction in the central processing unit (CPU) time by approximately three times less for the scattering simulation⁶⁴. The combination of the fast Fourier transform (FFT) and the conventional FMM, known as FMM-FFT, is also widely recognized^{65–68}. The FMM-FFT has been proven to minimize the computational complexity requirement of $O(N \log N)$ and $O(N^{4/3} \log^{2/3} N)$ in electromagnetic scattering from two

dimensional (2D) rough surface⁶⁵ and three-dimensional (3D) object scattering problems⁶⁶, respectively. Furthermore, numerous studies have performed FMM on the graphics processing unit (GPU) based parallel implementation to further accelerate the simulation time and handle larger scattering simulations with large unknowns^{68–72}.

The method proposed by He et al.⁷¹ utilizes the GPU-accelerated massively parallel multilevel FMM algorithm^{60,61,73,74} to solve extremely large scattering problems with 10 billion unknowns of electrically large and complicated objects, such as the ship model. To verify the performance of the parallel GPU and multilevel FMM (GPU-PMLFMA), an algorithm is implemented to extract the RCS from large spheres. In this process, the GPU-PMLFMA has high accuracy compared to the Mie series scattering. Furthermore, by using the GPU-PMLFMA, a significant speedup with respect to the number of unknowns in the ship model is reported (Fig. 3a).

Hybridization of the method of auxiliary sources (MAS) with the FMM has been employed to analyze the scattering problems of large arrays of circular dielectric cylinders (Fig. 3b)⁷⁵. After the simulation, the hybrid method exhibits a faster execution time than the standalone MAS method,

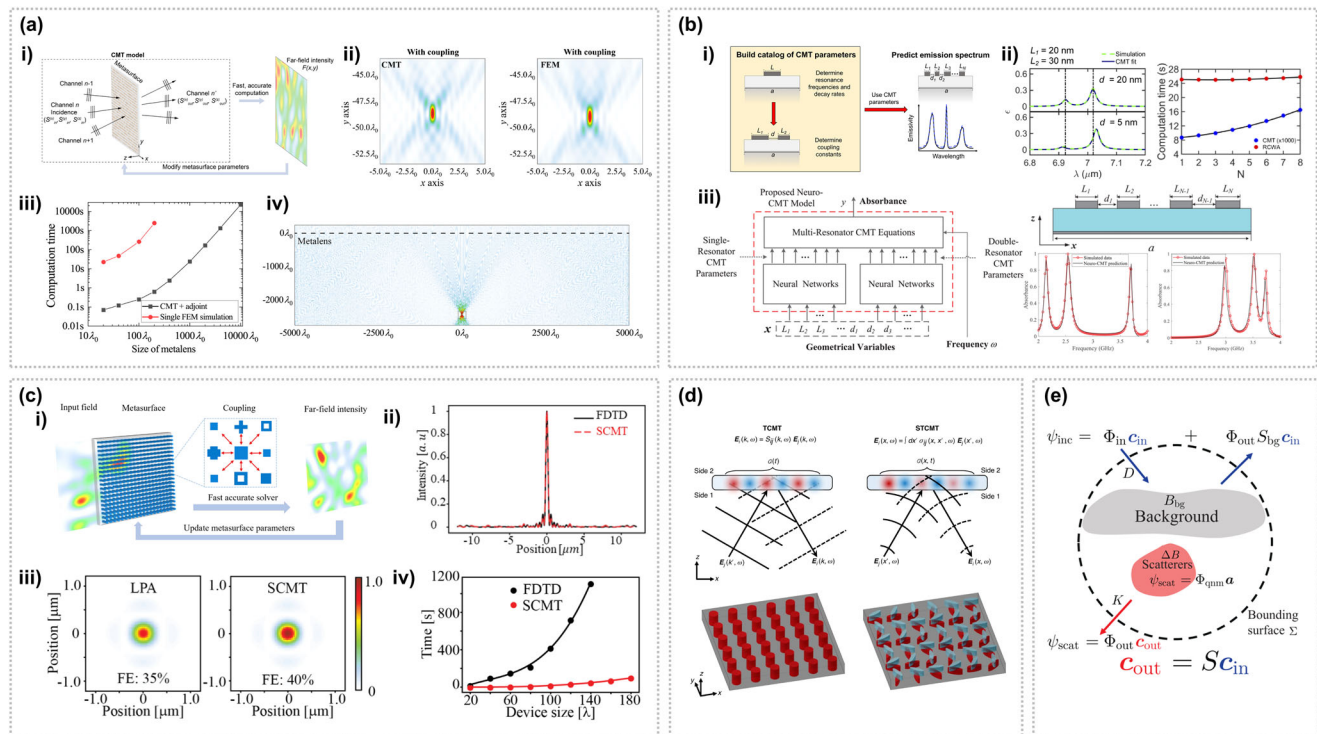


Fig. 4 | Fast simulations with coupled mode theories (CMTs). **a** Far-field intensity simulation of a metasurface using temporal coupled mode theory (TCMT)³². **a.i** Schematic of a large-area metasurface comprising M interconnected resonators. **a.ii** Comparison of normalized intensity profiles derived from the CMT and finite element method (FEM). **a.iii** Comparison of the computation times of CMT (black squares) and FEM simulations (red circles), achieving faster results by several orders of magnitude. **a.iv** Total field intensity from a metatens with a size of $10,000\lambda_0$. **b** Multi-resonance characteristic simulations using multi-resonance TCMT⁸⁷. **b.i** Emission spectrum of an infinite array of coupled hBN ribbons. **b.ii** Comparison of the computation times of the CMT (blue) and rigorous coupled-wave analysis

(RCWA; red). **b.iii** Structure and absorbance result of the neural network–combined CMT model. **c** Far-field intensity simulation of a metasurface using the spatial coupled mode theory (SCMT)⁹⁰. **c.i** Schematic of a large-area metasurface comprising interconnected waveguides. **c.ii** Point spread functions of the SCMT (red) and finite difference time domain (FDTD; black). **c.iii** Focal efficiency comparison designed by local periodic approximation and SCMT. **c.iv** Comparison of the computation times of the SCMT (red) and FDTD (black). **d** Nonlocal metasurfaces with spatially invariant meta-atoms using TCMT and those with spatially variant meta-atoms using spatiotemporal coupled mode theory⁷⁸. **e** Schematic of the quasi-normal coupled mode theory system⁸².

and the RCS results obtained using the hybrid method agree well with those of the COMSOL simulation.

Recently, the FMM has been implemented to solve 2D scattering problems of large-scale electromagnetic metasurfaces (Fig. 3c)³¹. The FMM is integrated with generalized sheet transition conditions (GSTCs) of the integral equations (IE) to reduce computational complexity. A simulation with a finite-sized metasurface is conducted using Gaussian beam illumination to compare the FMM IE-GSTC algorithm with a standard IE-GSTC solver. The results show that the FMM IE-GSTC can reduce the complexity to $O(N^{3/2})$ from $O(N^3)$ of the standard solver. This algorithm is subsequently applied to a large simulation domain. The total fields are computed using the FMM IE-GSTC in an electrically large two-room scenario with and without metasurfaces. In the presence of the three metasurfaces, the results show that the total field distribution is improved and covers blind areas.

Coupled mode theory

Coupled mode theory (CMT) is used to describe a system in which isolated waveguides and cavities are perturbed or interconnected⁷⁶. It has been extensively employed to investigate power transfer interactions between elements through coupling⁷⁷ and has served as an interpretive tool for comprehending the scattering behavior of intricate resonant electromagnetic systems. The essential components of CMT are as follows: (i) resonant modes of the system and (ii) ports that transmit energy to and from the resonant scatterer⁷⁸.

Temporal CMT (TCMT) explains the interaction between optical resonators and ports in a time-dependent system⁷⁹. This enables us to understand how energy is selectively transferred across ports based on the

frequency or how the spectral line shape varies⁸⁰. The key equations governing TCMT for a single resonant mode system are given by Eqs. (2) and (3):

$$\frac{d}{dt}a = (i\omega_0 - \gamma)a + \kappa^T s_+ \quad (2)$$

$$s_- = Cs_+ + da \quad (3)$$

Here, a represents the amplitude of the resonant mode with resonant frequency ω_0 and decay rate γ . s_+ and s_- are the incoming and outgoing wave amplitudes at the M ports coupled to the resonator with coupling coefficients given by the vectors κ and d respectively⁸¹. The direct scattering process between ports is given by the C matrix⁸². The characteristics of the TCMT have been established using diverse symmetry criteria inherent to the optical system⁸³. The three prevailing conditions are time-reversal symmetry, energy conservation, and Lorentz reciprocity. Optical systems that fulfill only one of these three limitations are frequently observed^{84,85}.

CMT can be used as a surrogate model along with an optimization technique to design a metasurface with the desired functionality³². A metasurface can be regarded as M interconnected resonators, which can be interpreted using TCMT (Fig. 4a). TCMT can be used when each resonator is lossless and only supports a single resonance. In this case, M resonators support the metasurface system, which is expressed as a M -dimensional column vector. Furthermore, ports can be defined as a discretized in-plane wave vector k_{xy} , with a spacing of $\Delta k = \frac{2\pi}{L}$. Using CMT guarantees a simulation without degradation in quality compared with the FEM. With a

substantial increase in the computational time, the design area is limited to a maximum of $200\lambda_0$ when utilizing the FEM simulation. However, in the case of the CMT, it can be increased to $10,000\lambda_0$.

TCMT can be utilized to model devices with complicated structures. To enhance the performance of metasurface devices, various component elements must be integrated into photonic structures. However, this leads to a significant expansion of the design space, resulting in an exponential increase in the computing cost⁸⁶. A previous study demonstrated that the use of TCMT in complicated, structured devices with multi-resonance frequencies can lead to a decrease in the calculation time compared with rigorous coupled-wave analysis (RCWA) (Fig. 4b)⁸⁷. The relationship between the CMT parameters and geometric design parameters in a system with more than N modes is stored in a lookup table⁸⁷. This table is generated by conducting full-wave electromagnetic simulations of subunit cells. This method requires additional full-wave simulations as new parameter values are added, which leads to the deterioration of efficiency. To resolve this issue, machine learning is integrated with CMT. This enables rapid prediction of the electromagnetic response of metasurfaces consisting of complex resonators⁸⁸.

However, a TCMT based on resonance may not be suitable for accurately simulating a metasurface dependent on the propagation of guided modes⁸⁹. Hence, the spatial CMT (SCMT)⁹⁰ is more appropriate for representing a metasurface as a collection of truncated waveguides. The key equation governing SCMT is given by Eq. (4):

$$C \frac{dU(z)}{dz} = i(BC + K)U(z) \quad (4)$$

where the waveguide mode amplitudes $U(z)$ obey a coupled differential equation involving the coupling matrices C and K and the propagation constant matrix B ⁹¹. Similar to TCMT, the mode profile is initially acquired for the isolated element, followed by an analysis of mutual coupling, leading to the formation of super modes and propagation constants⁹⁰. By employing SCMT, it is possible to obtain comparable outcomes to those of the FDTD method while significantly enhancing the computational efficiency; it can also be used to enhance the focal efficiency (FE) of the lens after using LPA during optimization (Fig. 4c)⁹⁰.

To expand the TCMT, spatiotemporal CMT (STCMT) has been proposed, which adds dependence on spatial dispersion. TCMT postulates that the magnitude of the resonator is solely determined as a function of time. This assumption is valid for finite periodic grating structures; however, TCMT may not be appropriate for finite and spatially varying gratings (Fig. 4d)⁷⁸. The governing equations accounting for spatial dispersion in STCMT are presented as Eqs. (5) and (6):

$$\frac{d}{dt}a(x, t) - i(\omega_0 + i\gamma_0)a(x, t) - (c + i\gamma_1)\frac{d}{dx}a(x, t) - i(b + i\gamma_2)\frac{d^2}{dx^2}a(x, t) = \int \kappa^*(x, x')s_+(x')dx' \quad (5)$$

$$s_-(x) = \int [C(x, x')s_+(x') + d(x, x')a(x)]dx' \quad (6)$$

obtained by Taylor expanding the resonant frequency (ω_0 , c , b) and decay rate (γ_0 , γ_1 , γ_2) in momentum k and substituting into the TCMT equations⁷⁸. Spectral-spatial features become significant when the optical energy propagates over a distance larger than the wavelength. In other words, these nonlocal or spatial dispersion characteristics are important for determining the response to the moment of light^{92,93}. Therefore, research on designing a metasurface using nonlocal characteristics is being conducted in various fields, such as augmented reality applications⁹⁴, and spectral⁹⁵ and chiral sensing⁹⁶. By expanding TCMT with SCMT, STCMT fully captures and demonstrates various aspects of nonlocal metasurface research, such as the resonant dynamics of numerous spectrally overlapping nonlocal modes, leaky-wave metasurfaces using traveling-wave q-BICs, and nonreciprocal behavior.

Finally, all aforementioned CMTs are constructed based on fundamental assumptions, including weak coupling and resonances with a high-quality (Q) factor⁷⁶. Large-area metasurfaces composed of several low Q, and highly coupled resonators may not be suitable for applying those methodologies^{97,98}. Because of the low Q of the resonator, CMT parameters are no longer frequency-independent. This conflicts with the underlying principles of the conventional CMT, thereby demanding interpretation within the framework of a more general theory, that is, the quasi-normal CMT (QCMT; Fig. 4e)⁸². The key QCMT equations are given by Eqs. (7)–(11):

$$iN(\omega)(\Omega - \omega)a = K^T(\omega)C_{in} \quad (7)$$

$$c_{out} = \left\{ C + \frac{i\omega}{4\alpha\beta^*} (\Phi_{inc}^{TR}, \Delta\Phi_{inc}) \right\} + D(\omega)a \quad (8)$$

$$K(\omega) = \frac{i\omega}{\alpha} (\Phi_{Lqnm}, \Delta B\Phi_{inc})^T \quad (9)$$

$$D(\omega) = \frac{i\omega}{4\beta^*} (\Phi_{inc}^{TR}, \Delta B\Phi_{Rqnm}) \quad (10)$$

$$S = C + \frac{i\omega}{4\alpha\beta^*} (\Phi_{inc}^{TR}, \Delta B\Phi_{inc}) - iD(\omega)[N(\omega)(\Omega - \omega)]^{-1}K^T(\omega) \quad (11)$$

Here, the coupling matrices K and D become frequency dependent. An additional term is also introduced in the scattering matrix equation to describe the Born scattering approximation ignored in conventional CMT⁸². From a computational perspective, one significant advantage of the QCMT is its ability to precisely resolve spectra around very high Q modes without the need for a dense, uniform frequency grid, which traditional frequency simulations require due to their lack of information on mode locations. For optical metasurfaces and other complex systems with high-Q modes, the QCMT approach can precisely and effectively resolve these modes, improving system design and analysis⁹⁹.

Neural network

Neural networks¹⁰⁰ are a type of machine learning model inspired by the structure and function of the human brain. They consist of layers of interconnected nodes, or “neurons,” which process input data to produce an output. Each connection between neurons has an associated weight that is adjusted during training to minimize the difference between the predicted and actual outputs. The basic unit of a neural network is the neuron, which receives input from other neurons, processes it using an activation function, and passes the result to subsequent neurons.

In the context of nanophotonics, neural networks are used for surrogate modeling, acting as approximations of more complex physical models. This approach is particularly useful because direct simulations of nanophotonic devices can be computationally expensive and time-consuming. Neural networks can be trained to quickly predict the behavior of these devices based on input variables, such as geometric parameters, material properties, and operating conditions. The input data is typically divided into training, validation, and test datasets^{101,102}. The training dataset is used to teach the network, the validation dataset helps tune the model and prevent overfitting, and the test dataset evaluates the model’s performance. Regular evaluation during training, such as using cross-validation, allows for early stopping if the model stops improving, thereby saving computational resources. By learning from datasets generated from simulations or experiments, neural networks efficiently map input parameters to output properties, making them powerful tools for rapid predictions and facilitating the design and optimization of nanophotonic devices and metasurfaces.

One method involves training a deep neural network (DNN) with a cylindrical meta-atom as the input and the phase and amplitude as the output. When learning a DNN, abrupt changes in the vicinity of the resonant frequency can pose challenges in achieving a balance between

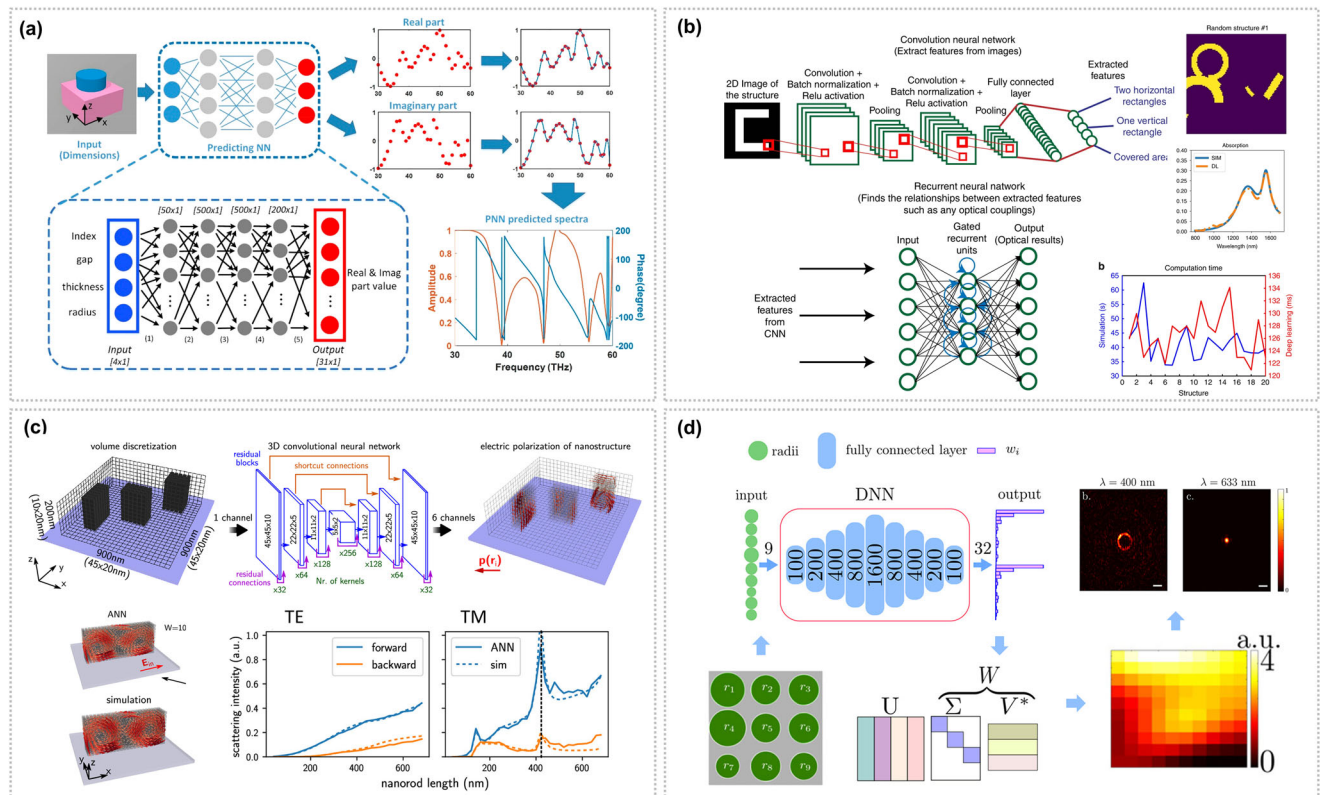


Fig. 5 | Fast simulation of optical metasurfaces assisted by neural networks.

a Deep neural network considers meta-atom characteristics, such as the index, gap, thickness, and radius as inputs, predicting the real and imaginary parts of the transmission coefficient. Amplitude and phase are then calculated separately based on these predictions¹⁰⁴. **b** Convolutional neural network captures spatial data from a 100×100 grid and uses a recurrent neural network to map these data to absorption

spectra¹⁰⁵. **c** A three dimensional (3D) convolutional neural network learns effectively from the input data of arbitrary 3D shapes, generating outputs represented by a 3D electric field³³. **d** Deep neural network trained with nine radius inputs implies an increase in the unit cell size and a reduction in the number of incorrect predictions of coupling with the surrounding meta-atoms¹⁰⁶.

overfitting and underfitting¹⁰³. In the study by An et al.¹⁰⁴, the real and imaginary parts of the transmission coefficient are predicted by varying the meta-atom characteristics, such as the index, gap, thickness, and radius (Fig. 5a). Subsequently, the amplitude and phase are calculated based on these predictions. More than 50,000 sets of randomly created 1×4 input vectors are fed into the bilinear tensor layer. Of these, 70% are allocated to the training set, and the remaining 30% are designated for the test set. The prediction of cylindrical and “H”-shaped meta-atoms shows a performance improvement of 600 times compared with the existing simulations, achieving 99% accuracy.

This deep learning-based method can also be extended to convolutional neural networks (CNNs) and recurrent neural networks (RNNs). CNNs use convolutional layers to extract spatial features from input data, typically images, whereas RNNs utilize recurrent connections to capture temporal dependencies in sequential data, such as time series or natural languages. In the study by Sajedian et al.¹⁰⁵, spatial information from a freeform pattern arranged in a 100×100 grid is extracted using a CNN and mapped to the absorption spectra using an RNN (Fig. 5b). To provide the data required for the model, they perform 100,000 simulations with comparable setups and random configurations. Sixty percent of this data is used to train the model, comprising the training dataset. Thirty percent is used to test the model, forming the test dataset. The remaining ten percent is utilized to validate the model, serving as the validation dataset. As illustrated in the figure, this model demonstrates a significantly faster performance than the existing simulations.

In particular, CNNs offer significant advantages in spatial feature extraction, making them increasingly promising for future use in the field of optics. The use of 3D convolution calculations enables learning

from an input with a 3D arbitrary shape and generates an output with a 3D electric field (Fig. 5c)³³. Both the input and output are represented as discretized voxels within the encoder–decoder framework. This approach demonstrates the ability to predict various physical quantities, including far-field scattering patterns, heat generation, and nonlinear effects, based on the calculated electric field values. In both cases of plasmonic and dielectric structures, they use 28,000 samples for training purposes and the remaining 2000 samples for validation and benchmarking. This method proves to be effective for both plasmonic and dielectric structures, with a failure rate of less than 10% and a computational speed of three to five times faster than that of simulations performed using the existing CPUs.

Although numerous studies have focused on predicting the optical response at the meta-atom level, some have also considered the coupling between meta-atoms (Fig. 5d)¹⁰⁶. The objective of analyzing the coupling between meta-atoms is to develop a simulation method that is more accurate than LPA based methods, while being faster and more memory-efficient than traditional grid-based methods. The input parameters for the DNN include not only the radius of the corresponding unit cell, as in the study by An et al.¹⁰⁴, but also the radii of the eight adjacent unit cells, totaling nine radii. This was done to create a simulation model that considers the coupling with adjacent unit cells. The DNN produces a weight matrix that separates the real and imaginary parts of the electric field. The dimensions of the weight matrix are reduced using singular value decomposition (SVD). They extract a random subset of 98,568 columns as training data to fit the matrix, while the remaining 24,642 columns are used for validation. Despite the error, the simulation using the trained model requires only 12 s, whereas the original FDTD simulation requires 3.1 h. In addition, the requirements

of random access memory for the initialization mesh (58.95 GB) and simulation (29.6 GB) are significantly reduced to 3.75 GB.

Notably, this modeling approach exhibits enhanced synergy when integrated with the inverse design method using neural networks, which will be explained in Section “Inverse design method”.

Inverse design method

In the typical forward design of metasurfaces, an optimal solution is obtained by defining the intrinsic physical properties such as phase maps, amplitude maps, or polarization distributions. Meta-atoms are then selected from libraries based on their similarity to these optimal solutions from human’s perspective^{7,107}. However, genuinely implementing an optimal solution with meta-atoms becomes increasingly challenging in the forward design process as the metasurface size increases, primarily owing to the growing degrees of freedom. Inverse designs have emerged to overcome the limitations inherent in previous forward design methodologies^{29,108}.

The desired function of metasurfaces in the inverse design is implemented based on computational algorithms. The inverse design maximizes or minimizes the desired function, called the figure of merit (FOM), by utilizing diverse methods such as topology optimization^{41,109}, heuristics¹¹⁰, and deep learning¹¹¹. In this process, human intuition is almost excluded and specific algorithms either imitate natural systems¹¹² or utilize the gradient¹¹³ as a guide for a proper FOM. The elimination of human intuition has traded the challenges posed by large complexities in terms of the advantages associated with the degrees of freedom, thus overcoming the constraints of a traditional forward design.

However, computational burden is a significant bottleneck in the inverse design process of aperiodic large-area metasurface. In heuristic-based methods, convergence is not guaranteed when associated with a large number of design parameters¹¹⁴. In traditional gradient-based optimization, the gradient calculation of several design variables consumes a considerable amount of time. Although achieving a fast-forward simulation is feasible, the challenge lies in the inability to update the design variables, rendering the task of designing large-area metasurfaces unattainable. To overcome these limitations, among various inverse design methods, the adjoint method and deep learning have been the most actively employed.

Adjoint-based optimization

In the adjoint method, the gradients of all design variables are computed using only two simulations with Green’s function symmetry and Lorentz reciprocity¹¹⁵. The adjoint method substitutes the design variables with each dipole moment. In the optimization process, the variations of these dipole moments create changes in the electric (or magnetic) fields, denoted as $\delta E(x_0)$ (or $\delta H(x_0)$) at the specified target location x_0 and these changes can be expressed as $G(x_0, x')p^{ind}(x')$. Here, $G(x_0, x')$ is Green’s function at a target point x_0 from a dipole of unit amplitude at x' and $p^{ind}(x')$ is the induced dipole moment in the design region x' . Without using the adjoint method, N iterations of iterative simulations are required to calculate $G(x_0, x')$ for each N variable, imposing a significant computational burden on the optimization process of numerous design variables. However, in the adjoint method, $G(x_0, x')$ can be replaced with $G(x', x_0)$, and Green’s function at individual x' from the adjoint source at x_0 is ensured by Green’s function symmetry and the Lorentz reciprocity theorem. Consequently, all $G(x', x_0)$ of each variable can be simultaneously computed using one adjoint simulation. This reduces the number of iterations of the total simulations by a factor of two, enabling the optimization of large-complexity problems.

Due to the fast gradient calculation of the adjoint method, it can be applied from the freeform optimization of meta-atoms to problems related to large areas of metasurfaces, where effective control over numerous design variables is essential. In the freeform structure, the meta-grating designed using the adjoint method exhibits superior diffraction efficiency, particularly at high angles (Fig. 6a)⁴¹. Periodic meta-atoms with freeform shapes can generate many more nano-optical modes, referred to as Bloch modes than their counterparts with simpler structures. This allows the creation of a more

diverse set of out-coupled modes in the desired diffraction channel. The strong interference between these modes results in high-efficiency diffraction, surpassing the capabilities of traditional meta-gratings. Furthermore, under periodic conditions, other functionalities as well as waveguides have been demonstrated, such as angle-tunable birefringence in optics and diverse applications in nonlinear optics, including second-harmonic generation^{109,116}. Under aperiodic conditions, the multi-functional feasibility of the adjoint method has been demonstrated in the study by Chung et al.³⁵ with high-NA achromatic metalens (Fig. 6b). Minimax optimization has been implemented with a freeform structure to maximize the minimum intensity between the diverse frequencies. Although the designed metalens has a 2D freeform structure that is impossible to fabricate, it has demonstrated achromatic capabilities with a high minimum efficiency (23%) at a high NA (0.99), through inverse design.

The advantages of the adjoint method can be effectively exploited for large-area designs. Large-area metasurfaces often entail numerous design sections that must be optimized, presenting a more complex optimization task than smaller-area metasurfaces. The efficiency of the adjoint method is particularly valuable in managing and optimizing numerous design variables across expansive metasurface configurations.

However, simulation times related to the meshes can still lead to problems in terms of computational cost. To address this issue, solutions such as utilizing cylindrical symmetry (Fig. 6c) or restricting the meta-atom structure to a specific subspace (Fig. 6d), such as rectangular shapes, have been proposed. The use of cylindrical symmetry reduces the simulation dimensions from 3D to 2D, significantly alleviating the cost of forward simulations over large areas¹¹⁷. In addition, constraining the structures of meta-atoms can mitigate mesh sensitivity¹¹⁸. This method restricts the structure of meta-atoms to parameterized rectangles in which all gradients are computed simultaneously using adjoint optimization. This work demonstrates a robust convergence to subwavelength variations, even at low resolutions. Both of these methods, through full FDTD simulations, address the drawbacks of unit cell-based metasurfaces caused by LPA.

Recently, centimeter-scale chromatic aberration-corrected metalens has been developed for practical applications such as VR (Fig. 6e)⁹. The metalens, composed of rectangular-specific subspace meta-atoms, minimizes simulation costs by employing cylindrical symmetry and a custom-made fast forward simulator. The dimensional gradients of every pixel’s meta-atom are calculated with only two simulations using the adjoint method, further minimizing the computational costs. Experimental validations with an NA of 0.7 at the millimeter scale and 0.3 at the centimeter scale have illustrated the feasibility of multifunctionality and the robustness of abrupt phase changes in large areas. This also has demonstrated its practical application in VR imaging systems with achromatic functionality at red, green and blue wavelengths. Consequently, they have shown the potential for large-area metasurface design through the combination of the adjoint method and a fast-forward simulator.

Deep learning

As described in Section “Fast simulation method”, deep learning has been highly successful in forward modeling in terms of designing and optimizing metasurfaces. The algorithms demonstrated proficiency in learning such complex relationships between metasurface structures and their optical responses. In addition to its outstanding applications in forward modeling, deep learning can be effectively employed to perform the inverse design of metasurfaces.

Deep learning-based inverse design entails the reverse process of forward modeling; that is, it considers optical responses as an input and provides parameters as an output. Although forward design deals with one-to-one mapping between structures and optical responses, inverse modeling must handle the nonunique nature of the problem (one-to-many), where distinct designs can yield identical optical responses. To address this problem, a tandem strategy involves the sequential use of two neural networks: forward neural networks (FNNs) and inverse neural networks (INN). The FNN uses geometric parameters as inputs and provides optical responses as

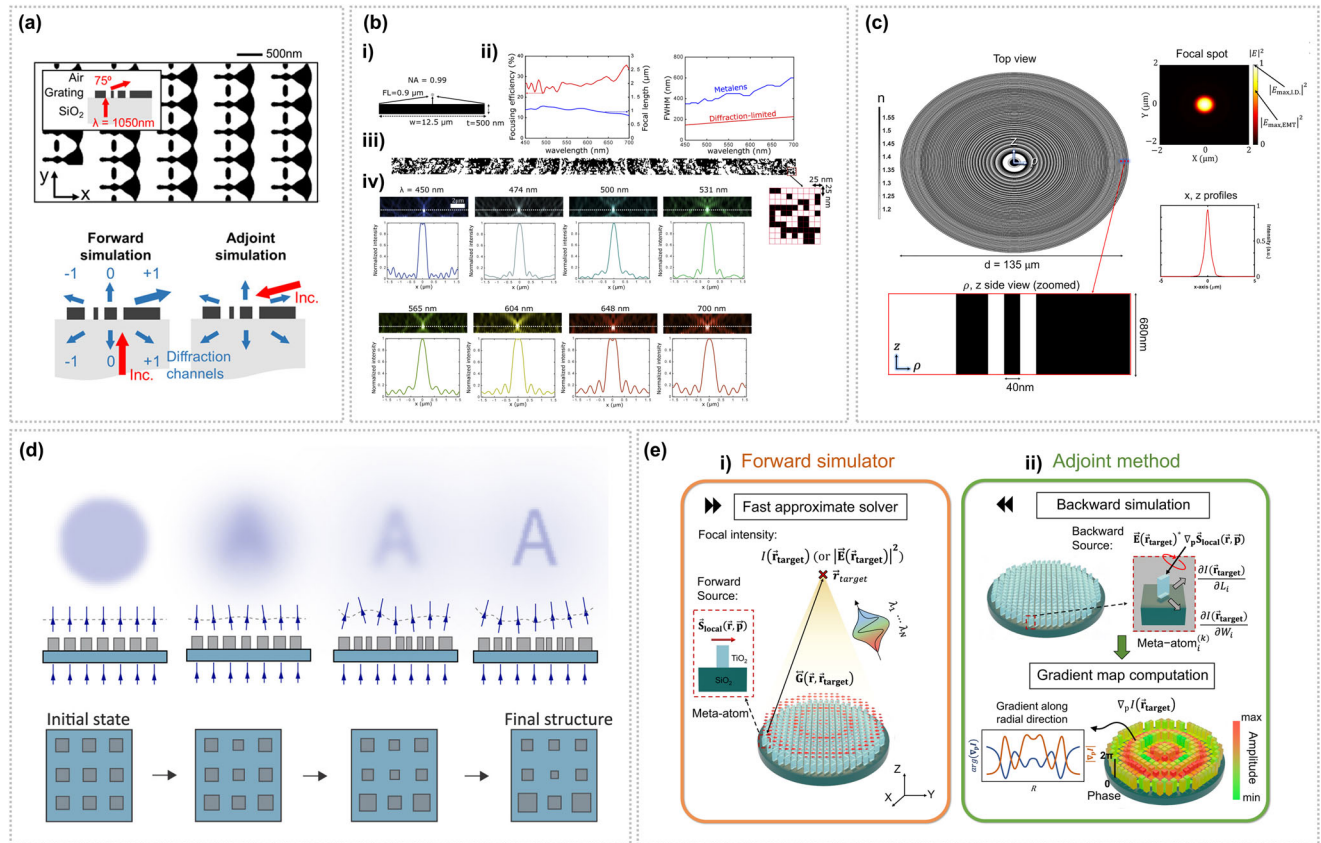


Fig. 6 | Inversely designed optical metasurfaces with the adjoint method.

a Freeform shape of the meta-atom under periodic condition for a high-performance waveguide⁴¹. **b** Inverse design-based, an achromatic metalens with extremely high numerical aperture (0.99)³⁵. **b.i** Schematic of the metalens. **b.ii** Performance of the metalens in the simulation. **b.iii** Shape of the metasurface, consisting of a non-fabricable two dimensional structure. **b.iv** Point spread function of the metalens. **c** Schematic of

cylindrical symmetric metalens¹¹⁷. **d** Design process of the metasurface with the meta-atom consisting of a specific subspace structure¹¹⁸. **e** Optimization schematic of a large-area metasurface⁹. **e.i** Forward simulator based on Chebyshev interpolation to conduct fast forward simulation. **e.ii** Backward simulation for calculating the gradients of the entire meta-atoms simultaneously with the adjoint method.

outputs. Once trained, the weights and biases of the FNN are frozen, and the INN is then connected to the FNN, which receives the optical responses outputted by the FNN and provides geometric parameters. Thus, the entire cascaded network works to solve one-to-one problems; hence, convergence is easier. In many studies, tandem-architecture neural networks have been utilized for inverse design in nanophotonics^{34,119–122}. In the study by Yeung et al.¹²⁰, the use of the tandem strategy has made it possible to design metal-insulator-metal supercell metasurfaces that were quadrilaterally symmetric (Fig. 7a). The tandem model performs a mapping between the absorption responses and design parameters.

Neural networks often exhibit a notable decrease in performance for data points far from the training set^{123,124}. Optimization techniques, such as particle swarm optimization and the adjoint method, are significantly influenced by the initial positions; therefore, they may fail in extreme cases¹²⁵. To address these limitations, hybrid algorithms have been investigated^{126,127}. Among the hybrid algorithms, the fusion of adjoint optimization and a generative adversarial network (GAN) in a single platform further improves the design results (Fig. 7b)¹²⁵. The obtained optical responses are passed through pre-trained neural networks, and the GAN then provides geometrical parameters. Finally, adjoint optimization is employed on the given designs to achieve superior designs provided by the GAN.

Reinforcement learning (RL) is useful when a labeled dataset is difficult to obtain. The RL algorithms learn by interacting with the environment and receiving feedback in the form of rewards and penalties. Deep Q-networks (DQNs) further advance RL by employing neural networks to handle

complex scenarios, showcasing the synergy between deep learning and RL¹²⁸. Its superior performance in the nanophotonic field has been proved^{129,130}. The DQN developed by Seo et al.¹³⁰ enables the design and optimization of a one-dimensional Si metasurface (Fig. 7c). The value of each cell can be either +1 for Si and −1 for air, and DQN is used to take actions among the N cells toggling its material between Si and air. It shows a high deflection efficiency (98.4%) of the meta-grating optimal structure compared with the previously studied optimization result¹¹⁵ and the electric field contribution of the optimized device.

We explored optimization methods employing neural networks, in which both training and inference rely on data-driven techniques. However, larger design areas entail higher computational costs for data collection, which is a significant drawback for data-driven neural networks. Physics-informed neural networks (PINNs) offer solutions by incorporating known physical laws into the training process. This integration of physical intuition helps improve efficiency and minimizes the reliance on exhaustive datasets. Various researchers have used PINNs to design and optimize metasurfaces^{36,131,132}. In particular, PINN optimize large-area metalens in the study by Zhelyeznyakov et al.³⁶ without relying on the LPA method. Their PINN operated without the need for a training dataset, that is, the training involved randomly generating distributions of dielectric meta-atoms, inputting them into the neural network and minimizing the residual of the linear Maxwell partial differential equation operators (Fig. 7d). After training, the PINN-optimized lens is compared with a traditional LPA lens. The results show that both the intensity and overall efficiency of the optimized lens are significantly improved.

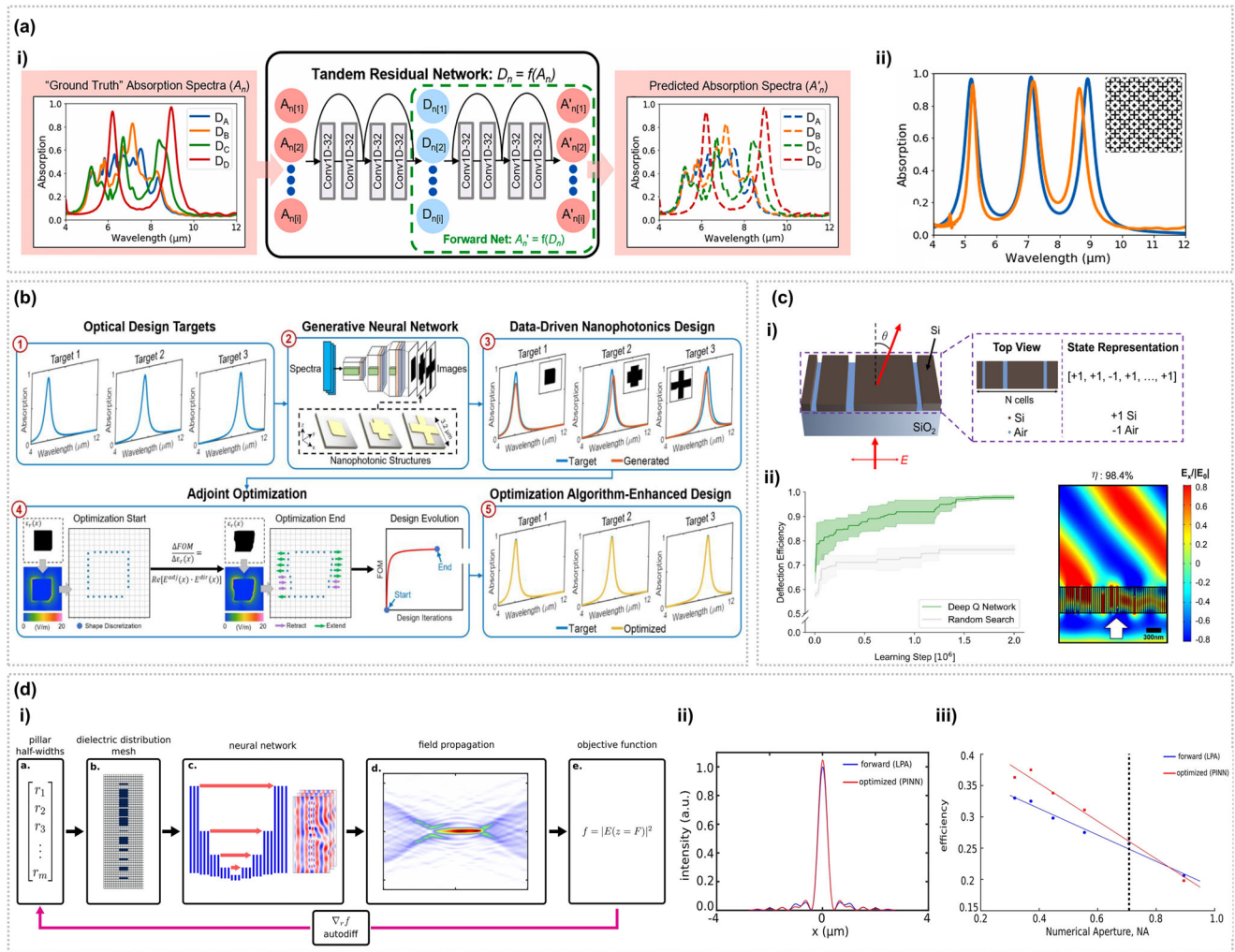


Fig. 7 | Inversely designed optical metasurfaces with deep learning. **a** A tandem strategy involves two neural networks, forward neural networks and inverse neural networks¹²⁰. **a.i** Schematic of data preparation for the deep learning model. **a.ii** Examples of the inverse design of new supercell metasurfaces exhibiting triple-narrowband (top) and graybody behaviors (bottom). **b** Schematic of the inverse design framework implementing the generative model with the adjoint-optimization method (DeepAdjoint)¹²⁵. **c** Deep Q-networks model for nanophotonics¹³⁰. **c.i** Schematic of a one-dimensional Si metagrating structure that splits into $N = 64$ cells with $+1$ or -1 representing Si and air, respectively.

c.ii Optimization results of the Si metagrating beam deflector. Deflection efficiencies during the overall training with the deep Q network (green) and random search (gray) with the darker line indicating the mean value (left) and electric field distribution of the optimized device (right). **d** Physics-informed neural networks (PINN) model for large-area metasurfaces³⁶. **d.i** Optimization framework of two dimensional meta-optics using PINNs. **d.ii** Comparison of intensity profiles from lenses designed using local phase approximation and PINNs. **d.iii** Efficiencies of both lenses versus the numerical aperture.

Conclusions

In conclusion, computational problems arising from the design of large-area metasurfaces can be effectively mitigated by employing rapid, accurate simulation methods and effective optimization tools. Various modified boundary conditions have been devised for the accurate and rapid simulations of large-area metasurfaces. The FMM expedites the computation of multiple poles across diverse sub-domains through hierarchical partitioning, significantly reducing the calculation times for determining the optical responses in expansive areas via local expansion. In the CMT, the coupling between adjacent nanostructures is computed by evaluating several resonant modes of the system and combining them, thereby facilitating the rapid and accurate calculation of optical responses by considering the coupling effects. Moreover, neural network-based simulators offer a viable alternative to conventional EM simulations, demonstrating significantly improved speeds and acceptable levels of accuracy, particularly after training appropriate optical responses.

In optimization, the performance limitation of a forward design with a large size can be compensated through inverse design methodologies.

Especially, optimization techniques based on the adjoint method and deep learning simplify the calculation process of gradients, enabling the optimization of large and complex problems^{115,133}. Additionally, these methods demonstrate that metasurfaces designed through inverse design exhibit superior performance compared to those designed through forward design approaches in large complexity problem hard to address with only human intuition⁹.

However, some challenges related to performance remain in scaling up metasurface. Although current metalenses at the millimeter to centimeter scale have demonstrated potential for practical imaging applications, there remains a significant efficiency gap compared with ROEs^{10,134}. To overcome this limitation, it is helpful not only to enhance the performance of the metasurface but also to implement techniques such as image reconstruction with neural networks^{135,136}. Furthermore, for the practical utilization of metasurfaces in the industry, the cost and time of fabrication are crucial factors. As a result, scalable fabrication methods, such as the roll to roll process or nanoimprinting, are required, which naturally introduce an increasing number of fabrication errors. Performance degradation owing to

fabrication errors is more critical for metasurfaces with insufficient average efficiency compared with ROEs¹³⁷. Therefore, for large-area metasurfaces to be effectively utilized in future devices, it is necessary to further develop optimization methods that not only ensure accurate and rapid design processes but also consider fabrication errors in design process while guaranteeing manufacturability using techniques such as morphology or deep learning^{138,139}.

Regardless of these limitations, various fabrication methods have been developed to produce large-area metasurfaces cost-effectively, and the combination of fast simulation techniques and inverse design has proven to be a high-performance approach to support large-area design. In addition, current open-source software and codes for nanophotonics, such as MEEP¹⁴⁰ and MetaBox¹³⁴, facilitate easy access to the simulation and optimization of large areas. The integration of optimization algorithms with commercial software such as MetaOptic Designer¹⁴¹ will further simplify the implementation of multifunctional metalenses on a large scale. These accessible tools enhance the potential for unrestricted design, pushing the boundaries of metasurface applications beyond size constraints. Consequently, it is anticipated that metasurfaces will surpass size limitations and become next-generation optical devices replacing ROEs in future optical platforms.

Received: 11 April 2024; Accepted: 19 June 2024;

Published online: 30 August 2024

References

- Khorasaninejad, M. et al. Metalenses at visible wavelengths: Diffraction-limited focusing and subwavelength resolution imaging. *Science* **352**, 1190–1194 (2016).
- Ni, X., Kildishev, A. V. & Shalaev, V. M. Metasurface holograms for visible light. *Nat. Commun.* **4**, 2807 (2013).
- Zhao, Y., Belkin, M. A. & Alù, A. Twisted optical metamaterials for planarized ultrathin broadband circular polarizers. *Nat. Commun.* **3**, 870 (2012).
- Pfeiffer, C. & Grbic, A. Metamaterial Huygens' surfaces: tailoring wave fronts with reflectionless sheets. *Phys. Rev. Lett.* **110**, 197401 (2013).
- Rubin, N. A. et al. Matrix Fourier optics enables a compact full-Stokes polarization camera. *Science* **365**, eaax1839 (2019).
- Li, Z. et al. Full-space cloud of random points with a scrambling metasurface. *Light Sci. Appl.* **7**, 63 (2018).
- Chen, W. T. et al. A broadband achromatic metalens for focusing and imaging in the visible. *Nat. Nanotechnol.* **13**, 220–226 (2018).
- Arbabi, A., Arbabi, E., Horie, Y., Kamali, S. M. & Faraon, A. Planar metasurface retroreflector. *Nat. Photon.* **11**, 415–420 (2017).
- Li, Z. et al. Inverse design enables large-scale high-performance meta-optics reshaping virtual reality. *Nat. Commun.* **13**, 2409 (2022).
- Li, Z. et al. Meta-optics achieves RGB-achromatic focusing for virtual reality. *Sci. Adv.* **7**, eabe4458 (2021).
- Song, W. et al. Ultraviolet metasurface-assisted photoacoustic microscopy with great enhancement in DOF for fast histology imaging. *Photoacoustics* **32**, 100525 (2023).
- Pahlevaninezhad, H. et al. Nano-optic endoscope for high-resolution optical coherence tomography in vivo. *Nat. Photon.* **12**, 540–547 (2018).
- Khorasaninejad, M. et al. Multispectral chiral imaging with a metalens. *Nano Lett.* **16**, 4595–4600 (2016).
- Jung, C. et al. Metasurface-driven optically variable devices. *Chem. Rev.* **121**, 13013–13050 (2021).
- Daqiqeh Rezaei, S. et al. Tri-functional metasurface enhanced with a physically unclonable function. *Mater. Today* **62**, 51–61 (2023).
- Gu, J. et al. Structural colors based on diamond metasurface for information encryption. *Adv. Opt. Mater.* **11**, 2202826 (2023).
- Dong, Z. et al. Schrödinger's red pixel by quasi-bound-states-in-the-continuum. *Sci. Adv.* **8**, eabm4512 (2022).
- Wang, H. et al. Two-photon polymerization lithography for imaging optics. *Int. J. Extreme Manuf.* **6**, 042002 (2024).
- Wang, H. et al. Two-photon polymerization lithography for optics and photonics: fundamentals, materials, technologies, and applications. *Adv. Funct. Mater.* **33**, 2214211 (2023).
- Kuznetsov, A. I. et al. Roadmap for optical metasurfaces. *ACS Photonics* **11**, 816–865 (2024).
- Pan, C.-F. et al. 3D-printed multilayer structures for high-numerical aperture achromatic metalenses. *Sci. Adv.* **9**, ead9262 (2023).
- Moon, S.-W. et al. Wafer-scale manufacturing of near-infrared metalenses. *Laser Photonics Rev.* 2300929 <https://doi.org/10.1002/lpor.202300929> (2024).
- Seong, J., Jeon, Y., Yang, Y., Badloe, T. & Rho, J. Cost-effective and environmentally friendly mass manufacturing of optical metasurfaces towards practical applications and commercialization. *Int. J. Precis. Eng. Manuf. Green Tech.* **11**, 685–706 (2024).
- Kim, J. et al. One-step printable platform for high-efficiency metasurfaces down to the deep-ultraviolet region. *Light Sci. Appl.* **12**, 68 (2023).
- Kim, J. et al. Scalable manufacturing of high-index atomic layer–polymer hybrid metasurfaces for metaphotonics in the visible. *Nat. Mater.* **22**, 474–481 (2023).
- Skarda, J. et al. Low-overhead distribution strategy for simulation and optimization of large-area metasurfaces. *npj Comput. Mater.* **8**, 78 (2022).
- Lin, Z., Liu, V., Pestourie, R. & Johnson, S. G. Topology optimization of freeform large-area metasurfaces. *Opt. Express* **27**, 15765–15775 (2019).
- Torfeh, M. & Arbabi, A. Modeling Metasurfaces Using Discrete-Space Impulse Response Technique. *ACS Photonics* **7**, 941–950 (2020).
- Li, Z., Pestourie, R., Lin, Z., Johnson, S. G. & Capasso, F. Empowering Metasurfaces with Inverse Design: Principles and Applications. *ACS Photonics* **9**, 2178–2192 (2022).
- Phan, T. et al. High-efficiency, large-area, topology-optimized metasurfaces. *Light Sci. Appl.* **8**, 48 (2019).
- Dugan, J., Smy, T. J. & Gupta, S. Accelerated IE-GSTC Solver for Large-Scale Metasurface Field Scattering Problems Using Fast Multipole Method (FMM). *IEEE Trans. Antennas Propag.* **70**, 9524–9533 (2022).
- Zhou, M. et al. Inverse design of metasurfaces based on coupled-mode theory and adjoint optimization. *ACS Photonics* **8**, 2265–2273 (2021).
- Wiecha, P. R. & Muskens, O. L. Deep learning meets nanophotonics: a generalized accurate predictor for near fields and far fields of arbitrary 3D nanostructures. *Nano Lett.* **20**, 329–338 (2020).
- So, S., Lee, D., Badloe, T. & Rho, J. Inverse design of ultra-narrowband selective thermal emitters designed by artificial neural networks. *Opt. Mater. Express OME* **11**, 1863–1873 (2021).
- Chung, H. & Miller, O. D. High-NA achromatic metalenses by inverse design. *Opt. Express* **28**, 6945–6965 (2020).
- Zhelyeznyakov, M. et al. Large area optimization of meta-lens via data-free machine learning. *Commun. Eng* **2**, 60 (2023).
- Pestourie, R. et al. Inverse design of large-area metasurfaces. *Opt. Express* **26**, 33732–33747 (2018).
- So, S., Mun, J., Park, J. & Rho, J. Revisiting the design strategies for metasurfaces: fundamental physics, optimization, and beyond. *Adv. Mater.* **35**, 2206399 (2023).
- Schubert, M. F. & Hammond, A. M. Fourier modal method for inverse design of metasurface-enhanced micro-LEDs. *Opt. Express* **31**, 42945–42960 (2023).
- Hsu, L., Dupré, M., Ndao, A., Yellowhair, J. & Kanté, B. Local phase method for designing and optimizing metasurface devices. *Opt. Express* **25**, 24974 (2017).

41. Sell, D., Yang, J., Doshay, S., Yang, R. & Fan, J. A. Large-angle, multifunctional metagratings based on freeform multimode geometries. *Nano Lett.* **17**, 3752–3757 (2017).
42. Chew, W. C. & Weedon, W. H. A 3D perfectly matched medium from modified maxwell's equations with stretched coordinates. *Microw. Opt. Technol. Lett.* **7**, 599–604 (1994).
43. Chew, W. C., Jin, J. M. & Michielssen, E. Complex coordinate stretching as a generalized absorbing boundary condition. *Microw. Opt. Technol. Lett.* **15**, 363–369 (1997).
44. Teixeira, F. L. & Chew, W. C. Differential forms, metrics, and the reflectionless absorption of electromagnetic waves. *J. Electromagn. Waves Appl.* **13**, 665–686 (1999).
45. Edee, K., Granet, G. & Plumey, J.-P. Complex coordinate implementation in the curvilinear coordinate method: application to plane-wave diffraction by nonperiodic rough surfaces. *J. Opt. Soc. Am. A: Opt. Image Sci. Vis.* **24**, 1097–1102 (2007).
46. Stratton, J. A. & Chu, L. J. Diffraction theory of electromagnetic waves. *Phys. Rev.* **56**, 99–107 (1939).
47. Lin, Z. & Johnson, S. G. Overlapping domains for topology optimization of large-area metasurfaces. *Opt. Express* **27**, 32445 (2019).
48. Greengard, L. *The rapid evaluation of potential fields in particle systems* (MIT Press, 1988).
49. Greengard, L. & Rokhlin, V. A fast algorithm for particle simulations. *J. Comput. Phys.* **73**, 325–348 (1987).
50. Ambrosiano, J., Greengard, L. & Rokhlin, V. The fast multipole method for gridless particle simulation. *Comput. Phys. Commun.* **48**, 117–125 (1988).
51. Greengard, L. & Gropp, W. D. A parallel version of the fast multipole method. *Comput. Math. Appl.* **20**, 63–71 (1990).
52. Shamshirgar, D. S., Yokota, R., Tornberg, A.-K. & Hess, B. Regularizing the fast multipole method for use in molecular simulation. *J. Chem. Phys.* **151**, 234113 (2019).
53. Kohnke, B., Kutzner, C. & Grubmüller, H. A GPU-Accelerated fast multipole method for GROMACS: performance and accuracy. *J. Chem. Theory Comput.* **16**, 6938–6949 (2020).
54. Ohno, Y. et al. Petascale molecular dynamics simulation using the fast multipole method on K computer. *Comput. Phys. Commun.* **185**, 2575–2585 (2014).
55. Yokota, R., Barba, L. A., Narumi, T. & Yasuoka, K. Petascale turbulence simulation using a highly parallel fast multipole method on GPUs. *Comput. Phys. Commun.* **184**, 445–455 (2013).
56. Dehnen, W. A fast multipole method for stellar dynamics. *Comput. Astrophys.* **1**, 1 (2014).
57. Ying, L. A pedestrian introduction to fast multipole methods. *Sci. China Math.* **55**, 1043–1051 (2012).
58. Huang, J., Jia, J. & Zhang, B. FMM-Yukawa: An adaptive fast multipole method for screened Coulomb interactions. *Comput. Phys. Commun.* **180**, 2331–2338 (2009).
59. Barnes, J. & Hut, P. A hierarchical O(N log N) force-calculation algorithm. *Nature* **324**, 446–449 (1986).
60. Song, J., Lu, C.-C. & Chew, W. C. Multilevel fast multipole algorithm for electromagnetic scattering by large complex objects. *IEEE Trans. Antennas Propag.* **45**, 1488–1493 (1997).
61. Sertel, K. & Volakis, J. L. Multilevel fast multipole method solution of volume integral equations using parametric geometry modeling. *IEEE Trans. Antennas Propag.* **52**, 1686–1692 (2004).
62. Coifman, R., Rokhlin, V. & Wandzura, S. The fast multipole method for the wave equation: a pedestrian prescription. *IEEE Antennas Propag. Mag.* **35**, 7–12 (1993).
63. Engheta, N., Murphy, W. D., Rokhlin, V. & Vassiliou, M. S. The fast multipole method (FMM) for electromagnetic scattering problems. *IEEE Trans. Antennas Propag.* **40**, 634–641 (1992).
64. Bindiganavale, S. S. & Volakis, J. L. A hybrid FE-FMM technique for electromagnetic scattering. *IEEE Trans. Antennas Propag.* **45**, 180–181 (1997).
65. Wagner, R. L., Song, J. & Chew, W. C. Monte Carlo simulation of electromagnetic scattering from two-dimensional random rough surfaces. *IEEE Trans. Antennas Propag.* **45**, 235–245 (1997).
66. Waltz, C., Sertel, K., Carr, M. A., Usner, B. C. & Volakis, J. L. Massively parallel fast multipole method solutions of large electromagnetic scattering problems. *IEEE Trans. Antennas Propag.* **55**, 1810–1816 (2007).
67. Taboada, J. M. et al. High scalability FMM-FFT electromagnetic solver for supercomputer systems. *IEEE Trans. Antennas Propag. Mag.* **51**, 20–28 (2009).
68. Dang, V., Nguyen, Q. M. & Kilic, O. GPU cluster implementation of FMM-FFT for large-scale electromagnetic problems. *IEEE Antennas Wirel. Propag. Lett.* **13**, 1259–1262 (2014).
69. Dang, V., Tran, N. & Kilic, O. Scalable fast multipole method for large-scale electromagnetic scattering problems on heterogeneous CPU-GPU clusters. *IEEE Antennas Wirel. Propag. Lett.* **15**, 1807–1810 (2016).
70. Guan, J., Yan, S. & Jin, J.-M. An OpenMP-CUDA implementation of multilevel fast multipole algorithm for electromagnetic simulation on multi-gpu computing systems. *IEEE Trans. Antennas Propag.* **61**, 3607–3616 (2013).
71. He, W.-J. et al. Solving electromagnetic scattering problems with tens of billions of unknowns using gpu accelerated massively parallel MLFMA. *IEEE Trans. Antennas Propag.* **70**, 5672–5682 (2022).
72. Nguyen, Q. M., Dang, V., Kilic, O. & El-Araby, E. Parallelizing fast multipole method for large-scale electromagnetic problems using gpu clusters. *IEEE Antennas Wirel. Propag. Lett.* **12**, 868–871 (2013).
73. Lu, C.-C. & Chew, W. C. A multilevel algorithm for solving a boundary integral equation of wave scattering. *Microw. Opt. Technol. Lett.* **7**, 466–470 (1994).
74. Borries, O., Meincke, P., Jørgensen, E. & Hansen, P. C. Multilevel fast multipole method for higher order discretizations. *IEEE Trans. Antennas Propag.* **62**, 4695–4705 (2014).
75. Mastorakis, E., Papakanellos, P. J., Anastassiou, H. T. & Tsitsas, N. L. Analysis of electromagnetic scattering from large arrays of cylinders via a hybrid of the Method of Auxiliary Sources (MAS) with the Fast Multipole Method (FMM). *Mathematics* **10**, 3211 (2022).
76. Joannopoulos, J. D., Johnson, S. G., Winn, J. N. & Meade, R. D. *Photonic Crystals: Molding the Flow of Light*, 2nd ed. (Princeton University Press, 2008).
77. Sharma, E. K., Nath, J. P. & Dhingra, N. Coupled mode theory and coupled mode photonic devices: A Review. *Asian J. Phys.* **30**, 0–00 (2021).
78. Overvig, A., Mann, S. A. & Alù, A. Spatio-temporal coupled mode theory for nonlocal metasurfaces. *Light Sci. Appl.* **13**, 28 (2024).
79. Haus, H. A. *Waves and Fields in Optoelectronics* (Prentice-Hall, 1984).
80. Suh, W., Wang, Z. & Fan, S. Temporal coupled-mode theory and the presence of non-orthogonal modes in lossless multimode cavities. *IEEE J. Quantum Electron.* **40**, 1511–1518 (2004).
81. Fan, S., Suh, W. & Joannopoulos, J. D. Temporal coupled-mode theory for the Fano resonance in optical resonators. *J. Opt. Soc. Am. A Opt. Image Sci. Vis.* **20**, 569–572 (2003).
82. Zhang, H. & Miller, O. D. Quasinormal Coupled Mode Theory. Preprint at <https://doi.org/10.48550/arXiv.2010.08650> (2020).
83. Zhao, Z., Guo, C. & Fan, S. Connection of temporal coupled-mode-theory formalisms for a resonant optical system and its time-reversal conjugate. *Phys. Rev. A* **99**, 033839 (2019).
84. Feng, L., El-Ganainy, R. & Ge, L. Non-Hermitian photonics based on parity-time symmetry. *Nat. Photon.* **11**, 752–762 (2017).
85. Longhi, S. PT-symmetric laser absorber. *Phys. Rev. A* **82**, 031801 (2010).
86. Hegde, R. S. Deep learning: a new tool for photonic nanostructure design. *Nanoscale Adv.* **2**, 1007–1023 (2020).

87. Audhkhasi, R., Zhao, B., Fan, S., Yu, Z. & Povinelli, M. L. Spectral emissivity modeling in multi-resonant systems using coupled-mode theory. *Opt. Express* **30**, 9463 (2022).
88. Zhang, J. et al. Physics-driven machine-learning approach incorporating temporal coupled mode theory for intelligent design of metasurfaces. *IEEE Trans. Microw. Theory Tech.* **71**, 2875–2887 (2023).
89. Gigli, C. et al. Fundamental limitations of Huygens' metasurfaces for optical beam shaping. *Laser Photonics Rev.* **15**, 2000448 (2021).
90. Wu, Z., Huang, X., Yu, N. & Yu, Z. Inverse design of dielectric metasurface by spatial coupled mode theory. Preprint at <https://arxiv.org/abs/2304.00603> (2023).
91. Hardy, A. & Streifer, W. Coupled modes of multiwaveguide systems and phased arrays. *J. Lightwave Technol.* **4**, 90–99 (1986).
92. Landau, L. D., Lifshitz, E. M. & Pitaevskii, L. P. *Electrodynamics of continuous media*, vol. 8. (Elsevier Science, 1995).
93. Agarwal, G. S., Pattanayak, D. N. & Wolf, E. Electromagnetic fields in spatially dispersive media*. *Phys. Rev. B* **8**, 1447 (1974).
94. Song, J. H., van de Groep, J., Kim, S. J. & Brongersma, M. L. Non-local metasurfaces for spectrally decoupled wavefront manipulation and eye tracking. *Nat. Nanotechnol.* **16**, 1224–1230 (2021).
95. Tittl, A. et al. Imaging-based molecular barcoding with pixelated dielectric metasurfaces. *Science* **360**, 1105–1109 (2018).
96. Hong, J. et al. Nonlocal metasurface for circularly polarized light detection. *Optica* **10**, 134 (2023).
97. Khorasaninejad, M. & Capasso, F. Metalenses: Versatile multifunctional photonic components. *Science* **358**, eaam8100 (2017).
98. Lallanne, P. & Chavel, P. Metalenses at visible wavelengths: past, present, perspectives. *Laser Photonics Rev.* **11**, 1600295 (2017).
99. Benzaouia, M., Joannopoulos, J. D., Johnson, S. G. & Karalis, A. Quasi-normal mode theory of the scattering matrix, enforcing fundamental constraints for truncated expansions. *Phys. Rev. Res.* **3**, 033228 (2021).
100. Jiang, J., Chen, M. & Fan, J. A. Deep neural networks for the evaluation and design of photonic devices. *Nat. Rev. Mater.* **6**, 679–700 (2021).
101. Ding, X. et al. Metasurface-based optical logic operators driven by diffractive neural networks. *Adv. Mater.* **36**, 2308993 (2024).
102. Trisno, J. et al. Applying machine learning to the optics of dielectric nanoblobs. *Adv. Photonics Res.* **1**, 2000068 (2020).
103. Yun, J., Kim, S., So, S., Kim, M. & Rho, J. Deep learning for topological photonics. *Adv. Phys. X* **7**, 2046156 (2022).
104. An, S. et al. A deep learning approach for objective-driven all-dielectric metasurface design. *ACS Photonics* **6**, 3196–3207 (2019).
105. Sajedian, I., Kim, J. & Rho, J. Finding the optical properties of plasmonic structures by image processing using a combination of convolutional neural networks and recurrent neural networks. *Microsyst. Nanoeng.* **5**, 27 (2019).
106. Zhelyeznyakov, M. V., Brunton, S. & Majumdar, A. Deep learning to accelerate scatterer-to-field mapping for inverse design of dielectric metasurfaces. *ACS Photonics* **8**, 481–488 (2021).
107. Ren, H. et al. Complex-amplitude metasurface-based orbital angular momentum holography in momentum space. *Nat. Nanotechnol.* **15**, 948–955 (2020).
108. So, S., Badloe, T., Noh, J., Bravo-Abad, J. & Rho, J. Deep learning enabled inverse design in nanophotonics. *Nanophotonics* **9**, 1041–1057 (2020).
109. Shi, Z. et al. Continuous angle-tunable birefringence with freeform metasurfaces for arbitrary polarization conversion. *Sci. adv.* **6**, eaba3367 (2020).
110. Chen, W. et al. Ultra-compact and low-loss silicon polarization beam splitter using a particle-swarm-optimized counter-tapered coupler. *Opt. Express* **28**, 30701–30709 (2020).
111. Peurifoy, J. et al. Nanophotonic particle simulation and inverse design using artificial neural networks. *Sci. adv.* **4**, eaar4206 (2018).
112. Jafar-Zanjani, S., Inampudi, S. & Mosallaei, H. Adaptive genetic algorithm for optical metasurfaces design. *Sci. Rep.* **8**, 11040 (2018).
113. So, S. et al. Multicolor and 3d holography generated by inverse-designed single-cell metasurfaces. *Adv. Mater.* **35**, 2208520 (2023).
114. Oldewage, E. T. et al. *The perils of particle swarm optimization in high dimensional problem spaces*. Ph.D. thesis (University of Pretoria, 2017).
115. Miller, O. D. *Photonic design: from fundamental solar cell physics to computational inverse design* (University of California, Berkeley, 2012).
116. Sitawarin, C., Jin, W., Lin, Z. & Rodriguez, A. W. Inverse-designed photonic fibers and metasurfaces for nonlinear frequency conversion. *Photonics Res.* **6**, B82–B89 (2018).
117. Chung, H., Zhang, F., Li, H., Miller, O. D. & Smith, H. I. Inverse design of high-NA metalens for maskless lithography. *Nanophotonics* **12**, 2371–2381 (2023).
118. Mansouree, M., McClung, A., Samudrala, S. & Arbabi, A. Large-scale parametrized metasurface design using adjoint optimization. *ACS Photonics* **8**, 455–463 (2021).
119. Liu, D., Tan, Y., Khoram, E. & Yu, Z. Training deep neural networks for the inverse design of nanophotonic structures. *ACS Photonics* **5**, 1365–1369 (2018).
120. Yeung, C. et al. Multiplexed supercell metasurface design and optimization with tandem residual networks. *Nanophotonics* **10**, 1133–1143 (2021).
121. Zhen, Z. et al. Realizing transmitted metasurface cloak by a tandem neural network. *Photonics Res.* **9**, B229–B235 (2021).
122. Lin, R., Zhai, Y., Xiong, C. & Li, X. Inverse design of plasmonic metasurfaces by convolutional neural network. *Opt. Lett.* **45**, 1362–1365 (2020).
123. Novak, R., Bahri, Y., Abolafia, D. A., Pennington, J. & Sohl-Dickstein, J. Sensitivity and generalization in neural networks: an empirical study. Preprint at <https://arxiv.org/abs/2010.08650> (2018).
124. Fort, S., Nowak, P. K., Jastrzebski, S. & Narayanan, S. Stiffness: A new perspective on generalization in neural networks. Preprint at <https://arxiv.org/abs/1901.09491> (2019).
125. Yeung, C., Pham, B., Tsai, R., Fountaine, K. T. & Raman, A. P. Deepadjoint: an all-in-one photonic inverse design framework integrating data-driven machine learning with optimization algorithms. *ACS Photonics* **10**, 884–891 (2023).
126. Wiecha, P. R., Arbouet, A., Girard, C. & Muskens, O. L. Deep learning in nano-photonics: inverse design and beyond. *Photonics Res.* **9**, B182 (2021).
127. Xu, Y., Zhang, X., Fu, Y. & Liu, Y. Interfacing photonics with artificial intelligence: an innovative design strategy for photonic structures and devices based on artificial neural networks. *Photonics Res.* **9**, B135 (2021).
128. Mnih, V. et al. Human-level control through deep reinforcement learning. *Nature* **518**, 529–533 (2015).
129. Sajedian, I., Lee, H. & Rho, J. Double-deep Q-learning to increase the efficiency of metasurface holograms. *Sci. Rep.* **9**, 10899 (2019).
130. Seo, D., Nam, D. W., Park, J., Park, C. Y. & Jang, M. S. Structural optimization of a one-dimensional freeform metagrating deflector via deep reinforcement learning. *ACS Photonics* **9**, 452–458 (2022).
131. Fang, Z. & Zhan, J. Deep physical informed neural networks for metamaterial design. *IEEE Access* **8**, 24506–24513 (2020).
132. Tanriover, I., Hadibrata, W. & Aydin, K. Physics-based approach for a neural networks enabled design of all-dielectric metasurfaces. *ACS Photonics* **7**, 1957–1964 (2020).
133. Chen, M. K., Liu, X., Sun, Y. & Tsai, D. P. Artificial intelligence in meta-optics. *Chem. Rev.* **122**, 15356–15413 (2022).
134. Huang, L. et al. Broadband thermal imaging using meta-optics. *Nat. Commun.* **15**, 1662 (2024).

135. Tseng, E. et al. Neural nano-optics for high-quality thin lens imaging. *Nat. Commun.* **12**, 6493 (2021).
136. Choi, E. et al. 360° structured light with learned metasurfaces. *Nat. Photon.* <https://doi.org/10.1038/s41566-024-01450-x> (2024).
137. Yang, Y. et al. Integrated metasurfaces for re-envisioning a near-future disruptive optical platform. *Light Sci. Appl.* **12**, 152 (2023).
138. Tanriover, I., Lee, D., Chen, W. & Aydin, K. Deep generative modeling and inverse design of manufacturable free-form dielectric metasurfaces. *ACS Photonics* **10**, 875–883 (2023).
139. Schubert, M. F., Cheung, A. K. C., Williamson, I. A. D., Spyra, A. & Alexander, D. H. Inverse design of photonic devices with strict foundry fabrication constraints. *ACS Photonics* **9**, 2327–2336 (2022).
140. Oskooi, A. F. et al. Meep: A flexible free-software package for electromagnetic simulations by the FDTD method. *Comput. Phys. Commun.* **181**, 687–702 (2010).
141. Xu, C. et al. Fully automated inverse design solution for metalenses/metamaterials. *Proc. SPIE* **12428**, 1242802 (2023).

Acknowledgements

This work was financially supported by the POSCO-POSTECH-RIST Convergence Research Center program funded by POSCO, and the National Research Foundation (NRF) grant (RS-2024-00356928) funded by the Ministry of Science and ICT (MSIT) of the Korean government. H.K. acknowledges the POSCO Asia fellowship, and the Yuhan foundation *New Ilhan* fellowship. H.Kim acknowledges the Asan Foundation Biomedical Science fellowship, and the Presidential Science fellowship funded by the MSIT of the Korean government.

Author contributions

J.R. conceived the idea and initiated the project. M.C. and J.S. organized contents of recent simulation method for large-area metasurface. J.P. and H.K. organized the structure of inverse design method. M.C., J.P., J.S. and H.K. mainly wrote the manuscript. H. Kim, J.Y. and J.S. were partially

involved in writing the manuscript. All authors confirmed the manuscript. J.R. guided the entire work.

Competing interests

Authors M.C., J.P., J.S., H.K., H.Kim, Y.Y. and J.Seong declare no financial or non-financial competing interests. Author J.R. serves as an Editor of this journal and had no role in the peer-review or decision to publish this manuscript. Author J.R. declares no financial competing interests.

Additional information

Correspondence and requests for materials should be addressed to Junsuk Rho.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2024, corrected publication 2025